



Mathematics Part 1 — Cbse Half-Yearly Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	According to the concepts of the number system, what does the union of rational and irrational numbers form? (A) The set of Imaginary Numbers (B) The continuous, unbroken Real Number Line (C) The set of Integers (D) The cyclic number system Answer: Correct Option: (B) ► Why? B. By uniting the rational and the irrational, we build the continuous, unbroken Real Number Line where every real physical measurement has a home. [Ch 3: The World of Numbers]	1
2	Assertion (A): The sample space for rolling a standard die is {1, 2, 3, 4, 5, 6}. Reason (R): The sample space is the set of all possible outcomes in an experiment. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true because the numbers 1 through 6 represent every possible result that can appear when a standard die is rolled. The reason correctly explains this by defining the sample space as the complete set of all such possible outcomes. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1



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3	A ball is dropped from a height of 24 feet. Each time it bounces, it reaches $\frac{3}{4}$ of its previous height. What is the maximum height attained after the second bounce? (A) 18 feet (B) 10.125 feet (C) 7.59375 feet (D) 13.5 feet Answer: Correct Option: (D) ► Why? D. The maximum height after the second bounce is $18 \times 0.75 = 13.5$ feet. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
4	Which of the following correctly defines an explicit formula for a sequence? (A) A rule that gives the value of a term using the values of previous terms (B) A rule that uses the term's position number n to calculate its value (C) A rule that finds the sum of all terms in the sequence (D) A rule that calculates the difference between consecutive terms Answer: Correct Option: (B) ► Why? B. An explicit formula is a rule that uses the term's position number, n, to calculate its value directly. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
5	What is a quadrilateral whose vertices are concyclic called? (A) Symmetric quadrilateral (B) Isosceles quadrilateral (C) Equilateral quadrilateral (D) Cyclic quadrilateral Answer: Correct Option: (D) ► Why? D. When the vertices of a quadrilateral are concyclic, the quadrilateral is called a cyclic quadrilateral. [Ch 5: I'm Up and Down, and Round and Round]	1
6	Which of the following are lines of reflection symmetry for a circle? (A) Only the horizontal and vertical diameters (B) Only the radii (C) All diameters (D) All chords Answer: Correct Option: (C) ► Why? C. All diameters are lines of reflection symmetry of the circle. [Ch 5: I'm Up and Down, and Round and Round]	1



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7	What is the coefficient of z in the polynomial $4z^3 + 5z^2 - 11$? (A) 5 (B) -11 (C) 0 (D) 4 Answer: Correct Option: (C) ► Why? C. The polynomial $4z^3 + 5z^2 - 11$ has no z term, meaning the coefficient of z is 0. [Ch 2: Introduction to Linear Polynomials]	1
8	Assertion (A): The numerator $x^2 - 7x + 12$ of a rational expression factors as $(x - 3)(x - 4)$. Reason (R): The denominator $5x^2 + 5x - 100$ of the same rational expression factors as $5(x - 4)(x + 5)$. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both the assertion and the reason are true statements regarding the factorization of the numerator and denominator. However, the factorization of the denominator does not explain why the numerator factors in a particular way. [Ch 4: Exploring Algebraic Identities]	1
9	If $\sqrt{2} = p/q$ leads to $2q^2 = p^2$, and we let $p = 2k$ (since p must be even), what is the resulting equation after substitution and simplification? (A) $q^2 = k^2$ (B) $2q^2 = 2k^2$ (C) $q^2 = 2k^2$ (D) $q^2 = 4k^2$ Answer: Correct Option: (C) ► Why? C. Substituting $p = 2k$ into $2q^2 = p^2$ gives $2q^2 = (2k)^2 = 4k^2$. Dividing both sides by 2 yields $q^2 = 2k^2$. [Ch 3: The World of Numbers]	1
10	Assertion (A): For an obtuse-angled triangle, the circumcentre lies inside the triangle. Reason (R): The circumcentre of a right-angled triangle lies at the midpoint of the hypotenuse. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	1



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	Answer: Correct Option: (D) ► Why? D. The assertion is false because for an obtuse-angled triangle, the circumcentre lies outside the triangle, not inside. The reason is a true statement about right-angled triangles. [Ch 5: I'm Up and Down, and Round and Round]	
11	Assertion (A): Brahmagupta's formula can be applied to any quadrilateral to find its area. Reason (R): Brahmagupta's formula generalises Heron's formula. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. The assertion is false because Brahmagupta's formula applies only to cyclic quadrilaterals, not to any quadrilateral. The reason is true because Brahmagupta's formula does generalise Heron's formula (by setting $d = 0$, a triangle becomes a special case of a cyclic 4-gon). [Ch 6: Measuring Space: Perimeter and Area]	1
12	What is the unit of measurement for area, based on a 1×1 square? (A) 1 sq. unit (B) 1 unit (C) 1 cubic unit (D) 1 perimeter unit Answer: Correct Option: (A) ► Why? A. For area, the unit is a 1×1 square; its area is taken to be 1 unit² (also written as 1 sq. unit). [Ch 6: Measuring Space: Perimeter and Area]	1
13	Assertion (A): A tree diagram is useful for visualising multi-step experiments where each path from start to end represents one complete outcome. Reason (R): A multi-step experiment involves a series of independent trials. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both A and R are true, but R is NOT the correct explanation of A. The assertion is true because a tree diagram visually maps out all paths from start to end as complete outcomes. The reason is true as it defines a multi-step experiment, but this definition does not explain the visual utility of the tree diagram.	1



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	<i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	
14	The Babylonians found through measurement that for a circle, $C^2/A \approx 12$. What formula for the area of a circle does this yield? (A) $A \approx C^2/12$ (B) $A \approx 12C$ (C) $A \approx C/12$ (D) $A \approx 12C^2$ Answer: Correct Option: (A) ► Why? A. If $C^2:A \approx 12:1$, then $A \approx C^2/12$. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
15	The straight lines representing an equation of the form $y = ax$ always pass through which point? (A) (0, 0) (B) (1, 1) (C) (a, 0) (D) (0, a) Answer: Correct Option: (A) ► Why? A. The straight lines representing an equation of the form $y = ax$ always pass through the origin (0, 0). <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
16	What does it mean for a coin to be considered 'fair' or 'unbiased'? (A) It is symmetrical so there is no reason for it to come down more often on one side. (B) It has heads on both sides. (C) It always lands on tails. (D) It is made of a specific metal. Answer: Correct Option: (A) ► Why? A. A fair or unbiased coin is symmetrical, meaning there is no reason for it to land on one side more often than the other. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



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17	What is the difference between an algebraic identity and an equation? (A) An identity is true for all values of the variables, while an equation need not be true for all values. (B) An equation is true for all values of the variables, while an identity is true for only one value. (C) Identities can only have one variable, while equations can have multiple variables. (D) There is no difference between an identity and an equation. Answer: Correct Option: (A) ► Why? A. An identity is an equation that is true for all values of the variables occurring in it, while an equation need not be true for all values. [Ch 4: Exploring Algebraic Identities]	1
18	Who formalised the notion and use of zero and negative numbers as algebraic entities, making the four-quadrant Cartesian plane possible? (A) Brahmagupta (B) René Descartes (C) Al-Bīrūnī (D) Ōmar Khayyām Answer: Correct Option: (A) ► Why? A. Brahmagupta formalised the notion and use of zero and negative numbers, which are essential for the negative axes and the origin in the four-quadrant Cartesian plane. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
19	What is the x-coordinate of a point that lies on the y-axis? (A) 0 (B) 1 (C) -1 (D) Undefined Answer: Correct Option: (A) ► Why? A. The x-coordinate of a point on the y-axis is 0. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
20	The letters of the word 'PEACE' are placed on cards, and one card is drawn without looking. What is the probability that the card is not an E? (A) $\frac{3}{5}$ (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) $\frac{4}{5}$ Answer: Correct Option: (A)	1



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	► Why? A. The word PEACE has 5 letters, with 2 E's. The letters that are not E are P, A, and C (3 letters). The probability is $\frac{3}{5}$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
Section B Attempt 5 out of 7.		
21	Describe the construction steps to locate $\sqrt{2}$ on the number line using a unit square's diagonal. Answer: Measure OA = 1 unit on the number line and draw a perpendicular at A. Mark point B such that AB = 1 unit, making OB = $\sqrt{2}$ units. With O as center and OB as radius, draw an arc intersecting the number line at P, representing $\sqrt{2}$. ► Why? This tests the geometric construction method described in the text, relying on the Baudhāyana-Pythagoras Theorem to create the length $\sqrt{2}$ and transferring it via a compass. [Ch 3: The World of Numbers]	2
	OR Find the area of a quadrant of a circle whose circumference is 44 cm. (Use $\pi = \frac{22}{7}$) Answer: Given $2\pi r = 44$, we find $r = 7$ cm. The area of a quadrant (a sector with 90° angle) is $\frac{1}{4} \pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 7^2 = 38.5 \text{ cm}^2$. ► Why? The radius is derived from the circumference, and the quadrant's area is found by taking one-fourth of the total circle area since its central angle is 90°. [Ch 6: Measuring Space: Perimeter and Area]	
22	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). Answer: relative frequency, 0.3 (or 15/50) ► Why? According to the provided context, the ratio of the number of times an event occurs to the total number of trials is called the relative frequency (e.g., relative frequency of rolling a 4 is $\frac{8}{50} = 0.16$). Here, the number of students who like football is 15 out of a total of 50 students. Thus, the relative frequency is calculated as $\frac{15}{50}$, which simplifies to 0.3. ► Step-by-step: - Step 1: Identify the term for the ratio of observed occurrences to total trials, which is relative frequency. - Step 2: Calculate the relative frequency using the formula: Relative frequency = (Number of students who like football) / (Total number of students) = $\frac{15}{50} = 0.3$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	2
	OR An arc of a circle subtends an angle of 110° at the centre. What is the measure of the angle subtended by this arc at a point on the circle outside the arc?	



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	Answer: The angle subtended by an arc at the centre is double the angle subtended by it at any point on the circle outside the arc. Therefore, the required angle is $\frac{1}{2} \times 110^\circ = 55^\circ$. ► Why? Theorem 9 states that the central angle is always twice the angle subtended at any point on the circle outside the arc. [Ch 5: I'm Up and Down, and Round and Round]	
23	Bela has ₹100 for pocket money and spends ₹5 every day. Write a linear expression for the amount of money left on the n^{th} day. Answer: The expression for the amount left on the n^{th} day is $100 - 5n$. This represents the initial amount minus the product of the daily expenditure and the number of days. ► Why? The expression models the remaining amount by subtracting the total amount spent ($5n$) from the initial amount (100). [Ch 2: Introduction to Linear Polynomials]	2
24	What is the side of the cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$ cubic units? Answer: The volume can be rewritten as $(p)^3 + 3(p)^2(2q) + 3(p)(2q)^2 + (2q)^3$, which matches the form $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. Thus, the side is $(p+2q)$ units. ► Why? This tests the application of the cube identity to determine the dimensions of a geometric solid based on its volume expression. [Ch 4: Exploring Algebraic Identities]	2
25	State the general formula for the distance between two points (x_1, y_1) and (x_2, y_2) in the 2-D plane. Answer: The distance between the points (x_1, y_1) and (x_2, y_2) is given by the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. This distance formula is derived using the Baudhāyana-Pythagoras Theorem. ► Why? The student must recall and state the general distance formula explicitly provided in the text, linking it to its foundational theorem. [Ch 1: Orienting Yourself: The Use of Coordinates]	2
Section C Attempt 5 out of 7.		
26	Describe the steps to construct a line segment of length $\sqrt{2}$ on the number line using a ruler and compass.	3



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	Answer: Step 1: On the number line, measure OA = 1 unit and draw a perpendicular on OA through point A. Step 2: On this perpendicular, mark point B such that AB = 1 unit. Join the origin O to B. By the Baudhāyana-Pythagoras Theorem, $OB^2 = OA^2 + AB^2 = 1 + 1 = 2$, so $OB = \sqrt{2}$ units. Step 3: With O as centre and OB as radius, draw an arc using a compass that intersects the number line at P. Then $OP = \sqrt{2}$ units, and P represents the irrational number $\sqrt{2}$ on the number line. ∴ Point P on the number line represents $\sqrt{2}$, constructed using the diagonal of a unit square. ► Why? The construction uses a 1-unit perpendicular to form a right triangle with hypotenuse $\sqrt{2}$, then transfers that length to the number line via an arc. ► Step-by-step: - Step 1: On the number line, mark $OA = 1$ unit and draw a perpendicular at A - Step 2: Mark $AB = 1$ unit on the perpendicular; join O to B so that $OB = \sqrt{2}$ by the theorem - Step 3: With O as centre and OB as radius, draw an arc intersecting the number line at P; then $OP = \sqrt{2}$ [Ch 3: The World of Numbers]	
	OR When $d = 0$ in a 4-gon with sides a, b, c, d, show that the semi-perimeter of the 4-gon equals the semi-perimeter of the triangle with sides a, b, c. Answer: Given: 4-gon with sides a, b, c, d where $d = 0$. Formula: Semi-perimeter of 4-gon = $\frac{1}{2}(a + b + c + d)$ Substitution: $s = \frac{1}{2}(a + b + c + 0)$ $s = \frac{1}{2}(a + b + c)$ ∴ The semi-perimeter of the 4-gon equals $\frac{1}{2}(a + b + c)$, which is exactly the semi-perimeter of the triangle with sides a, b, c. This equivalence is essential for showing that Brahmagupta's formula reduces to Heron's formula when $d = 0$, since both formulas share the same value of s. ► Why? Setting $d = 0$ removes the fourth side from the semi-perimeter calculation, making $\frac{1}{2}(a+b+c+d)$ equal to $\frac{1}{2}(a+b+c)$, the standard semi-perimeter of the triangle. ► Step-by-step: - Step 1: Write the semi-perimeter of the 4-gon: $s = \frac{1}{2}(a + b + c + d)$ - Step 2: Substitute $d = 0$ - Step 3: Simplify to get $s = \frac{1}{2}(a + b + c)$ - Step 4: Conclude this equals the triangle's semi-perimeter [Ch 6: Measuring Space: Perimeter and Area]	



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27	Find the value of the quadratic polynomial $7s^2 - 4s + 6$ when $s = -3$. Answer: Given: The quadratic polynomial is $7s^2 - 4s + 6$, and the value of $s = -3$. Formula: Substitute the value of the variable into the expression and apply the correct order of operations. Substitution: Value = $7(-3)^2 - 4(-3) + 6$ Calculation: = $7(9) + 12 + 6$ = $63 + 12 + 6$ = $75 + 6$ = 81 ∴ The value of the quadratic polynomial at $s = -3$ is 81. ► Why? Substituting a negative value into a quadratic polynomial requires careful application of exponent rules and sign conventions. ► Step-by-step: - Step 1: Substitute $s = -3$ into the expression $7s^2 - 4s + 6$. - Step 2: Evaluate the exponent $(-3)^2 = 9$, then multiply by 7 to get 63. - Step 3: Multiply -4 by -3 to get +12. - Step 4: Add the terms $63 + 12 + 6$ to get the final answer 81. [Ch 2: Introduction to Linear Polynomials]	3
	OR When triangle ADM is reflected in the y-axis, the image of point A(3, 4) is A' and the image of point D(7, 1) is D'. Calculate the distance A'D' using the coordinates of the reflected points.	



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	Answer: Given: Reflected points are A'(-3, 4) and D'(-7, 1). Formula: Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Substitution: $x_2 - x_1 = -3 - (-7) = 4$ $y_2 - y_1 = 4 - 1 = 3$ Calculation: $A'D' = \sqrt{4^2 + 3^2}$ $A'D' = \sqrt{16 + 9}$ $A'D' = \sqrt{25}$ $A'D' = 5$ units ∴ The distance A'D' is 5 units, which matches the original length AD since reflection preserves lengths. ► Why? The solution applies the distance formula to the reflected coordinates A' and D', showing that the calculated distance remains 5 units. ► Step-by-step: 1. Step 1: Identify the coordinates of the reflected points A'(-3, 4) and D'(-7, 1). 2. Step 2: Calculate the shifts along the axes: x-shift = -3 - (-7) = 4, y-shift = 4 - 1 = 3. 3. Step 3: Apply the Baudhāyana-Pythagoras Theorem: $A'D' = \sqrt{4^2 + 3^2} = 5$ units. [Ch 1: Orienting Yourself: The Use of Coordinates]	
28	Define an event in the context of probability. If a standard 6-sided die is rolled once, write the event for 'rolling a number greater than 4' and list its elements. Answer: Definition: An event is any single possible result or combination of results that might happen. It is like choosing particular outcomes from all that could occur, and it is a subset of a sample space. Given: Rolling a standard 6-sided die. Sample Space S = {1, 2, 3, 4, 5, 6}. Event: 'Rolling a number greater than 4' means the outcome must be 5 or 6. Elements: E = {5, 6} Answer: ∴ An event is a subset of a sample space, and the elements of this event are {5, 6}.	3



Q.NO.	QUESTIONS	MARKS
	► Why? An event represents a specific condition or outcome of interest from the broader sample space, containing only those results that satisfy the condition. ► Step-by-step: - Step 1: Define an event as a subset of the sample space representing specific outcomes. - Step 2: Identify the sample space for a 6-sided die: $S = \{1, 2, 3, 4, 5, 6\}$. - Step 3: Select the outcomes greater than 4 to form the event $E = \{5, 6\}$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
29	An arc AFB of a circle with centre C subtends an angle of 120° at the centre. Find the angle subtended by this arc at a point D on the circle outside the arc. Answer: Given: Angle subtended by arc AFB at centre C = 120° Theorem: The angle subtended by an arc at the centre is double the angle subtended by it at any point on the circle outside the arc. Formula: $\angle ACB = 2 \angle ADB$ Substitution: $120^\circ = 2 \angle ADB$ Calculation: $\angle ADB = 120^\circ / 2 = 60^\circ$ $\therefore$ The angle subtended by the arc at point D is 60°. ► Why? According to Theorem 9, the central angle is exactly twice the angle subtended by the same arc at any point on the circle outside the arc. ► Step-by-step: - Step 1: Identify the given angle at the centre (120°) and the unknown angle at the circumference. - Step 2: Apply Theorem 9: the angle at the centre is double the angle at the circumference outside the arc. - Step 3: Divide the central angle by 2 to find the angle at D: $120^\circ / 2 = 60^\circ$. [Ch 5: I'm Up and Down, and Round and Round]	3
30	Prove that for any three consecutive square numbers $(n-1)^2$, n^2, and $(n+1)^2$, the sum of the smallest and largest squares minus two times the middle square always equals 2. Answer: Given: Three consecutive squares $(n-1)^2$, n^2, and $(n+1)^2$. Formula: Sum of smallest and largest is $(n-1)^2 + (n+1)^2$. Calculation: $(n-1)^2 + (n+1)^2 = n^2 - 2n + 1 + n^2 + 2n + 1 = 2n^2 + 2$ Subtracting $2n^2$ (which is twice the middle square) from this sum: $2n^2 + 2 - 2n^2 = 2$ $\therefore$ The result is always 2.	3



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	► Why? Expanding the squares using identities and simplifying cancels the linear terms, leaving a constant difference of 2. ► Step-by-step: - Step 1: Expand the smallest square $(n-1)^2$ and largest square $(n+1)^2$ using the identities $(a-b)^2$ and $(a+b)^2$. - Step 2: Add the two expanded expressions to get $2n^2 + 2$. - Step 3: Subtract $2n^2$ from the sum to arrive at the constant value 2. [Ch 4: Exploring Algebraic Identities]	
Section D Attempt 4 out of 6.		
31	Brahmagupta introduced the concepts of "fortunes" (Dhana) and "debts" (Riña) to represent positive and negative numbers. Using these real-world concepts, explain why: (i) the product of two debts is a fortune, and (ii) subtracting a negative number is the same as adding a positive number. Provide a numerical example for each. Answer: Brahmagupta grounded his mathematics in the reality of commerce by introducing fortunes (Dhana) to represent positive numbers (wealth or assets) and debts (Riña) to represent negative numbers (debts). This real-world connection helps explain the rules of integer arithmetic logically. (i) The product of two debts is a fortune: A negative number represents a debt, and multiplying by a negative number represents the removal of that debt. If someone takes away four of your debts, each worth ₹3 (which is -3), you are effectively ₹12 richer. Mathematically, taking away a debt is multiplying by a negative: $(-3) \times (-4) = +12$ The removal of a debt increases your wealth, hence a fortune. (ii) Subtracting a negative number is the same as adding a positive number: If you have a wealth of ₹10 and someone takes away a debt of ₹5 (represented as -5), your net wealth increases. The removal of a -5 debt means you gain ₹5. This is expressed as: $10 - (-5) = 15$ Subtracting a debt adds to your fortune. Thus, Brahmagupta's rules perfectly mirror real-life financial transactions, showing that arithmetic operations on integers are logical and consistent.	5



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	► Why? The question requires interpreting the abstract rules of integer multiplication and subtraction through Brahmagupta's real-world analogies of wealth (fortunes) and loans (debts). ► Step-by-step: 1. Step 1: Define Brahmagupta's concepts of fortunes (positive) and debts (negative). 2. Step 2: Explain that multiplying two negatives means removing a debt, which increases wealth (product is positive). Give the numerical example $(-3) \times (-4) = 12$. 3. Step 3: Explain that subtracting a negative means taking away a debt, which adds to wealth (same as adding a positive). Give the numerical example $10 - (-5) = 15$. 4. Step 4: Conclude by summarizing how these rules reflect real-world commerce. [Ch 3: The World of Numbers]	
	OR Explain how the area of a circle formula $A = \pi r^2$ can be derived by slicing the circle and rearranging the slices into a parallelogram-like shape. Also, state how Archimedes expressed the area of a circle. Answer: Archimedes expressed the area of a circle as follows: "The area of a circle is equal to the area of a right angle triangle with sides equal to the radius and the circumference of the circle." This gives: Area of circle = $\frac{1}{2} \times \text{circumference} \times \text{radius} = \frac{1}{2} \times 2\pi r \times r = \pi r^2$ His proof was based on a property of regular polygons: the area enclosed by a regular polygon equals half the perimeter $\times$ the radius of the circle that fits tightly within the polygon. He asked what happens as the number of sides gets larger and larger, arriving at: $A = \pi r^2$ Nilakanṭha Somayājī's visual explanation (c. 1500): The circle is sliced into many thin sectors. These slices are rearranged alternately (up-down-up-down) to form a parallelogram-like structure. As the slices become smaller and smaller, the arcs become more and more closer to a straight line. This makes the figure more and more closer to a parallelogram with: • Base = half the circumference = πr (since the top and bottom each contribute half the total arcs) • Height = radius r Therefore, the area of the circle equals the area of this parallelogram: Area = base $\times$ height = $\pi r \times r = \pi r^2$ Thus, both the limiting argument of Archimedes and the slicing-rearrangement argument of Nilakanṭha Somayājī lead to the same formula $A = \pi r^2$.	



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	► Why? Slicing the circle into thin sectors and rearranging them yields a parallelogram with base = πr and height = r, giving area = πr^2. Archimedes used a limiting argument with regular polygons to reach the same result. ► Step-by-step: 1. Step 1: State Archimedes' expression: area = $\frac{1}{2} \times \text{circumference} \times \text{radius} = \frac{1}{2} \times 2\pi r \times r = \pi r^2$. 2. Step 2: Explain Archimedes' approach using regular polygons with increasing number of sides. 3. Step 3: Describe Nilakanṭha Somayājī's method: slice the circle into thin sectors and rearrange alternately. 4. Step 4: As slices become smaller, the figure approaches a parallelogram with base = πr and height = r. 5. Step 5: Compute area = base $\times$ height = $\pi r \times r = \pi r^2$. 6. Step 6: Conclude that both methods yield $A = \pi r^2$. [Ch 6: Measuring Space: Perimeter and Area]	
32	State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$.	5



Q.NO.	QUESTIONS	MARKS
	Answer: The identity for the square of the sum of three numbers is: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Geometric Interpretation: A square of side $(a + b + c)$ is drawn. The side is divided into three segments of lengths a, b, and c. Lines parallel to the sides are drawn at these division points, creating a grid of 9 regions inside the square. These regions consist of: • Three squares with areas a^2, b^2, and c^2. • Six rectangles — two each with areas ab, bc, and ca. The total area of the big square equals the sum of all these regions, giving the identity: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Finding 119^2: Given: $119 = 100 + 10 + 9$, so $a = 100$, $b = 10$, $c = 9$. Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $119^2 = (100 + 10 + 9)^2$ $= 100^2 + 10^2 + 9^2 + 2(100)(10) + 2(100)(9) + 2(10)(9)$ $= 10000 + 100 + 81 + 2000 + 1800 + 180$ $= 14161$ $\therefore 119^2 = 14161$ Expanding $(3x - 2y + 4z)^2$: Given: $a = 3x$, $b = -2y$, $c = 4z$ Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $(3x - 2y + 4z)^2 = (3x)^2 + (-2y)^2 + (4z)^2 + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x)$ $= 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$	



Q.NO.	QUESTIONS	MARKS
	$\therefore (3x - 2y + 4z)^2 = 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$ ► Why? The identity $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ arises from partitioning a square of side $(a+b+c)$ into 3 smaller squares and 6 rectangles, and it simplifies both numerical squaring and algebraic expansion. ► Step-by-step: Step 1: State the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Step 2: Describe the geometric model — a square of side $(a+b+c)$ divided into 3 squares and 6 rectangles, and label each region with its area. Step 3: Apply the identity to find 119^2 by expressing 119 as $100 + 10 + 9$ and substituting into the formula. Step 4: Expand $(3x - 2y + 4z)^2$ by identifying $a = 3x$, $b = -2y$, $c = 4z$ and carefully substituting while tracking signs. Step 5: Simplify all terms and state the final expanded expression. [Ch 4: Exploring Algebraic Identities]	
	OR The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in the circle. Find the distance of the chord from the centre. Also, a cyclic quadrilateral ABCD is inscribed in this circle with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find the measures of $\angle C$ and $\angle D$.	



Q.NO.	QUESTIONS	MARKS
	Answer: Part (a): Distance of chord from the centre Given: Diameter = 26 cm, Chord length = 24 cm Radius: $r = 26/2 = 13 \text{ cm}$ Property used: A perpendicular from the centre to a chord bisects the chord. Half of chord: $24/2 = 12 \text{ cm}$ Right triangle formed by radius, half-chord, and distance from centre: $d^2 + 12^2 = 13^2$ $d^2 + 144 = 169$ $d^2 = 169 - 144 = 25$ $d = \sqrt{25} = 5 \text{ cm}$ $\therefore$ Distance from centre to chord = 5 cm Part (b): Angles of cyclic quadrilateral ABCD Given: $\angle A = 75^\circ$, $\angle B = 110^\circ$, ABCD is a cyclic quadrilateral Property used: In a cyclic quadrilateral, opposite angles sum to 180°. Finding $\angle C$: $\angle A + \angle C = 180^\circ$ $75^\circ + \angle C = 180^\circ$ $\angle C = 180^\circ - 75^\circ = 105^\circ$ $\therefore \angle C = 105^\circ$ Finding $\angle D$: $\angle B + \angle D = 180^\circ$ $110^\circ + \angle D = 180^\circ$	



Q.NO.	QUESTIONS	MARKS
	$\angle D = 180^\circ - 110^\circ = 70^\circ$ $\therefore \angle D = 70^\circ$ Verification: Sum of all four angles = $75^\circ + 110^\circ + 105^\circ + 70^\circ = 360^\circ \checkmark$ Diagram description: Draw a circle with centre O. Draw chord of length 24 cm and show the perpendicular from O bisecting it. Mark the radius (13 cm), half-chord (12 cm), and distance (5 cm). Separately, draw cyclic quadrilateral ABCD inscribed in a circle, labelling all four angles. ► Why? The perpendicular from the centre bisects the chord, creating a right triangle with the radius, half-chord, and distance from centre, solved using the Pythagorean theorem. Opposite angles of a cyclic quadrilateral are supplementary (sum to 180°). ► Step-by-step: Step 1: Find the radius from the diameter: $r = 26/2 = 13$ cm. Step 2: Use the property that perpendicular from centre bisects the chord; half-chord = $24/2 = 12$ cm. Step 3: Apply the Pythagorean theorem in the right triangle: $d^2 + 12^2 = 13^2$, solve for $d = 5$ cm. Step 4: For the cyclic quadrilateral, use the property that opposite angles sum to 180° to find $\angle C = 105^\circ$ and $\angle D = 70^\circ$. Step 5: Verify that all four angles sum to 360° and describe the diagram. [Ch 5: I'm Up and Down, and Round and Round]	
33	A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). Answer: (i) Draw two perpendicular lines intersecting at the center (Origin). Using 1 cm = 200 m, draw 5 parallel lines at 1 cm intervals on both sides of the N-S road and 5 parallel lines at 1 cm intervals on both sides of the E-W road.	5



Q.NO.	QUESTIONS	MARKS
	► Why? The city model represents a 2-D Cartesian coordinate system where the main roads act as the x-axis and y-axis, crossing at the origin (0,0). Because distances to the right and upwards are positive, and distances to the left and downwards are negative, a specific street number (like the 4th street) exists on both the positive and negative sides of the origin. Therefore, any intersection (a, b) will have four possible locations in the grid: (a, b), (a, -b), (-a, b), (-a, -b) ► Step-by-step: 1. Step 1: Recognize the city model as a 2-D Cartesian coordinate system with the center crossing as the origin (0,0). 2. Step 2: Understand that N-S streets represent x-coordinates (positive to the East, negative to the West) and E-W streets represent y-coordinates (positive to the North, negative to the South). 3. Step 3: For intersection (4, 3), the 4th N-S street can be $x = 4$ or $x = -4$. The 3rd E-W street can be $y = 3$ or $y = -3$. The four possible intersections are: (4, 3), (4, -3), (-4, 3), (-4, -3) 4. Step 4: For intersection (3, 4), the 3rd N-S street can be $x = 3$ or $x = -3$. The 4th E-W street can be $y = 4$ or $y = -4$. The four possible intersections are: (3, 4), (3, -4), (-3, 4), (-3, -4) [Ch 1: Orienting Yourself: The Use of Coordinates]	
34	A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. Construct a table of values showing the total amount she will have at the end of the first 4 months. Find a linear expression to represent the amount she will have in the n^{th} month. Explain why this represents linear growth.	5



Q.NO.	QUESTIONS	MARKS										
	Answer: Given: Initial amount = ₹500. Monthly pocket money = ₹150. Calculation of Monthly Amounts: End of month 1: 500 + 1 × 150 = ₹650 End of month 2: 500 + 2 × 150 = ₹800 End of month 3: 500 + 3 × 150 = ₹950 End of month 4: 500 + 4 × 150 = ₹1100 Table of Values: <table><tr><td>Month (n)</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Amount (₹)</td><td>650</td><td>800</td><td>950</td><td>1100</td></tr></table> Linear Expression: The amount at the end of the nth month will be: ₹(500 + 150n) Explanation of Linear Growth: As the value of n increases by one month, the value of the total amount increases by a fixed amount of ₹150. Linear growth describes a linear pattern where a quantity increases by a constant amount over equal intervals. Therefore, this represents linear growth. ► Why? The total amount increases by a constant value of ₹150 for each additional month, forming an arithmetic progression. This constant increase over equal intervals characterizes a linear growth pattern. ► Step-by-step: 1. Step 1: Calculate the total amount at the end of the first 4 months by adding the initial amount to (number of months × 150). 2. Step 2: Construct a table of values showing the month number and the corresponding amount. 3. Step 3: Generalize the pattern to find a linear expression for the amount in the nth month. 4. Step 4: Explain that the expression represents linear growth because the amount increases by a constant value (₹150) over equal intervals (each month). [Ch 2: Introduction to Linear Polynomials]	Month (n)	1	2	3	4	Amount (₹)	650	800	950	1100	
Month (n)	1	2	3	4								
Amount (₹)	650	800	950	1100								



Q.NO.	QUESTIONS	MARKS
Section E Attempt 3 out of 4.		
35	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from $D_1 (9.5, 0)$ to $R_1 (11.5, 0)$. The bathroom door extends from $B_1 (0, 1.5)$ to $B_2 (0, 4)$. A rectangular study table has three of its feet at the points $(8, 9)$, $(11, 9)$ and $(11, 7)$. Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B_1 and B_2 have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? Answer: (i) Width = $11.5 - 9.5 = 2$ units (ii) $(8, 7)$. Since the table is rectangular, the fourth foot must have the same x-coordinate as $(8, 9)$ and the same y-coordinate as $(11, 7)$. (iii) Width of bathroom door = $4 - 1.5 = 2.5$ units. The bathroom door is wider than the room door ($2.5 > 2$). (iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero. ► Why? i) Width = $11.5 - 9.5 = 2$ units ii) $(8, 7)$. Since the table is rectangular, the fourth foot must have the same x-coordinate as $(8, 9)$ and the same y-coordinate as $(11, 7)$. iii) Width of bathroom door = $4 - 1.5 = 2.5$ units. The bathroom door is wider than the room door ($2.5 > 2$). iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero. [Ch 1: Orienting Yourself: The Use of Coordinates]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at (3, 2). A shop is located at (7, 2). Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. Answer: (i) (0, 0) (ii) The third quadrant (iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. (iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. ► Why? i) (0, 0) ii) The third quadrant iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. [Ch 1: Orienting Yourself: The Use of Coordinates]	
36	Case Study: Simplifying Rational Expressions In an algebra workshop, students learn to simplify rational expressions by factorising numerators and denominators. The teacher presents the expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$. To factorise $x^2 - 7x + 12$, students find two numbers with sum -7 and product 12, yielding $(x - 3)(x - 4)$. For the denominator, they first factor out 5 to get $5(x^2 + x - 20)$, then find two numbers with sum 1 and product -20, yielding $5(x - 4)(x + 5)$. Common factors are then cancelled since the denominator is not zero. (i) What two numbers have a sum of -7 and a product of 12? (ii) Factorise $5x^2 + 5x - 100$ completely. (iii) Simplify the rational expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$, assuming the denominator is not zero. (iv) Why can the factor $(x - 4)$ be cancelled during simplification of this rational expression?	5



Q.NO.	QUESTIONS	MARKS
	Answer: (i) -3 and -4 (since $-3 + (-4) = -7$ and $(-3)(-4) = 12$) (ii) $5x^2 + 5x - 100 = 5(x^2 + x - 20) = 5(x - 4)(x + 5)$ (iii) $(x^2 - 7x + 12)/(5x^2 + 5x - 100) = (x - 3)(x - 4)/[5(x - 4)(x + 5)] = (x - 3)/[5(x + 5)]$ (iv) The factor $(x - 4)$ is a common factor in both the numerator and denominator, and it can be cancelled because the denominator $5x^2 + 5x - 100$ is not equal to zero, ensuring $x - 4$ is not zero in this context. ► Why? i) -3 and -4 (since $-3 + (-4) = -7$ and $(-3)(-4) = 12$) ii) $5x^2 + 5x - 100 = 5(x^2 + x - 20) = 5(x - 4)(x + 5)$ iii) $(x^2 - 7x + 12)/(5x^2 + 5x - 100) = (x - 3)(x - 4)/[5(x - 4)(x + 5)] = (x - 3)/[5(x + 5)]$ iv) The factor $(x - 4)$ is a common factor in both the numerator and denominator, and it can be cancelled because the denominator $5x^2 + 5x - 100$ is not equal to zero, ensuring $x - 4$ is not zero in this context. [Ch 4: Exploring Algebraic Identities]	
37	Case Study: Monthly Billing and Linear Relationships A community centre charges a fixed annual fee and an additional cost per session for using the tennis court. A member observed that when she used the court for 8 sessions, her total bill was ₹600. When she used it for 12 sessions, her total bill was ₹840. The total bill y depends on the number of sessions x according to the linear relation $y = ax + b$. In this equation, a represents the slope of the line and b represents the y-intercept. The member wants to determine the values of a and b to understand the billing structure. (i) What does the value of 'a' represent in the equation $y = ax + b$? (ii) Find the values of a and b. (iii) At what point does the graph of this equation cut the y-axis? (iv) What does the value of b represent in the context of the community centre? Answer: (i) The value of 'a' represents the slope of the line. (ii) From the given information: $8a + b = 600$ and $12a + b = 840$. Subtracting: $4a = 240$, so $a = 60$. Substituting: $b = 600 - 480 = 120$. (iii) The graph cuts the y-axis at the point $(0, 120)$. (iv) The value $b = 120$ represents the fixed annual fee charged by the community centre, which is the y-intercept of the line. ► Why? i) The value of 'a' represents the slope of the line. ii) From the given information: $8a + b = 600$ and $12a + b = 840$. Subtracting: $4a = 240$, so $a = 60$. Substituting: $b = 600 - 480 = 120$. iii) The graph cuts the y-axis at the point $(0, 120)$. iv) The value $b = 120$ represents the fixed annual fee charged by the community centre, which is the y-intercept of the line. [Ch 2: Introduction to Linear Polynomials]	5

